

Quality and Reliability Evidence

Supporting Material

Supporting material: quality and reliability evidence

What can be checked immediately

Claim	Evidence
Components compile as standalone WASM tools	<code>demo/reproduce.sh</code> runs locked tests, host/wasm Clippy, and WASM release builds on the pinned Rust <code>1.96.1</code> toolchain.
Payment request validation and QR boundary	<code>solana-pay-request</code> README and evidence.
Final-byte-derived unsigned-transfer summary	<code>spl-transfer-build</code> README, especially its final-byte verification and external-signing workflow; evidence.
Invoice-bound, balance-delta confirmation	<code>solana-pay-confirm</code> README and evidence.
Independent adversarial review	<code>M5_SECURITY_AUDIT.md</code> : threat model, 24 attacks, findings, fixes, and post-remediation validation.
Plugin ABI / registry compatibility	<code>zeroclaw-plugins</code> README and vendored <code>wit/v0</code> .
Exact reusable low-level primitives	<code>nanosol</code> README and its Rust test suite.

Recorded external validation

- **Real ZeroClaw host:** all three components have host execution records using ZeroClaw `0.8.3` and WASM components, rather than only unit tests.
- **Real devnet builder acceptance:** `spl-transfer-build` evidence records an externally signed, public devnet transaction, independent byte decode, simulation, and final balance change. The component had only the sender's public key.
- **Real mainnet read-only checks:** builder evidence documents a mainnet simulation and a Token-2022 refusal. Confirmation evidence documents a real host call against mainnet and offline verification over captured raw transaction bytes.
- **Cross-plugin binding:** request and confirmation assert the same frozen, independently derived reference vector, while the builder attaches the same reference convention to transfer instructions.
- **Mutation resistance:** the builder evidence documents independently applied mutations of amounts, accounts, programs, flags, instructions, signatures, lookup data, and trailing bytes; each is refused rather

than paired with the old approval summary.

Limits disclosed rather than papered over

- A first-party request→paid→confirm `paid:true` recording is not already preserved. The existing devnet fixture was pruned and a new faucet attempt was rate-limited. The video SOP requires a newly funded disposable devnet payment before making that claim on camera.
- Confirmation depends on its operator-selected RPC. A second independently operated endpoint reduces a single-provider forged-positive risk but does not eliminate all external trust.
- The components are stateless; a merchant system must own delivery policy, duplicate-notification suppression, persistence, retries, and daily limits.
- The design is intentionally narrow: no native SOL confirmation, arbitrary instructions, swaps, generic signing, submission, or key custody.

Source navigation

Public repository locations (also linked in the write-up):

- [Plugin repository \(pinned source\)](#)
- [Shared core \(pinned source\)](#)
- [Draft upstream PR](#)
- [Solana Pay transfer-request specification](#)